

Implementation of a database management system for Research Project, Publication and Grant Management System

DBMS Course Project Report



Project Information	Details
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Project Type	DBMS Course Project
Programme	BTech AIML

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1. Abstract

The Research Project, Publication and Grant Management System is a relational database design that centralizes the information generated by academic research activity. It connects departments, researchers, research projects, investigators, funding agencies, grants, milestones, expenditure, publications, patents and reports in a structured model.

The design replaces scattered files with a single source of truth, so that project progress, funding and research outputs can be viewed, tracked and reported together. Relational constraints (primary keys and foreign keys) preserve the links between records and protect data integrity and consistency.

2. Introduction and Problem Statement

2.1 Introduction

Research work produces a large amount of related data: researchers belong to departments, work on projects, projects are funded by grants from agencies, progress is measured through milestones, money is spent against budgets, and the work results in publications, patents and reports. When these records are kept in separate files and systems, it becomes hard to see the complete picture. This project models the full research lifecycle in one relational database.

2.2 Problem Statement

Research-related information is often maintained across separate files and systems. This leads to the following problems:

- There is no single view of overall project progress.
- Grant and funding details may be scattered across records.
- Project milestones and expenditure are difficult to track.
- Manual data management can lead to duplication, inconsistency and missing information.

2.3 Proposed Solution

A centralized relational database system that connects researchers, projects, funding, milestones, expenditure and research outputs, so that every record is stored once and linked to the records it depends on.

3. Objectives and Scope

3.1 Objectives

- Centralize research project-related information.

- Maintain researcher and department records.
- Manage projects and investigator assignments.
- Track grants, funding agencies and budgets.
- Monitor project milestones and expenditure.
- Link projects with publications and patents.
- Generate meaningful research and funding reports.
- Maintain data integrity and consistency through relational database design.

3.2 Scope

The project covers the following functions:

- Register and maintain researcher, department and project details.
- Assign investigators to research projects.
- Manage funding agencies, grants and project budgets.
- Track project milestones and completion status.
- Record and monitor project expenditure and budget utilization.
- Manage research publications, patents and reports.
- Retrieve research information and generate project, funding and research-output reports.

4. System Users and Requirements

4.1 System Users

User	Responsibilities
Researcher	View assigned projects; track project milestones; edit research outputs; access publication information
Administrator	Register projects and researchers; assign investigators; manage grants and funding; record milestones; generate reports
Department	Monitor, track and review the research outputs

4.2 Functional Requirements

The system should allow authorized users to:

- Register researchers and departments.
- Create and manage research projects.
- Assign investigators to projects.
- Register funding agencies.
- Record grants and project budgets.
- Define and track milestones.

- Record project expenditure.
- Link publications to projects.
- Maintain patent information.
- Generate research and funding reports.

4.3 Software and Hardware Requirements

Category	Requirement
Operating System	Windows / macOS / Linux
Database	Relational DBMS (e.g., MySQL)
Database Client MySQL Workbench	Database Client MySQL Workbench
Query Language	SQL
Modelling Tools	ER diagram and schema design tool
Programming Language	Python 3.12
GUI Library	HTML5, CSS3, Vanilla JavaScript (ES6+)
Web Framework	Flask 3.1
Diagramming / Slides	Canva / PowerPoint
Processor	Intel Core i3
RAM	8 GB

5. ER Diagram

The conceptual ER diagram represents eleven entities and the relationships between them. A department has many researchers, and a researcher is assigned as an investigator. Investigators investigate projects (many-to-many). A project receives grants provided by funding agencies, and a project has milestones, expenditure records, reports, and produces publications and patents.

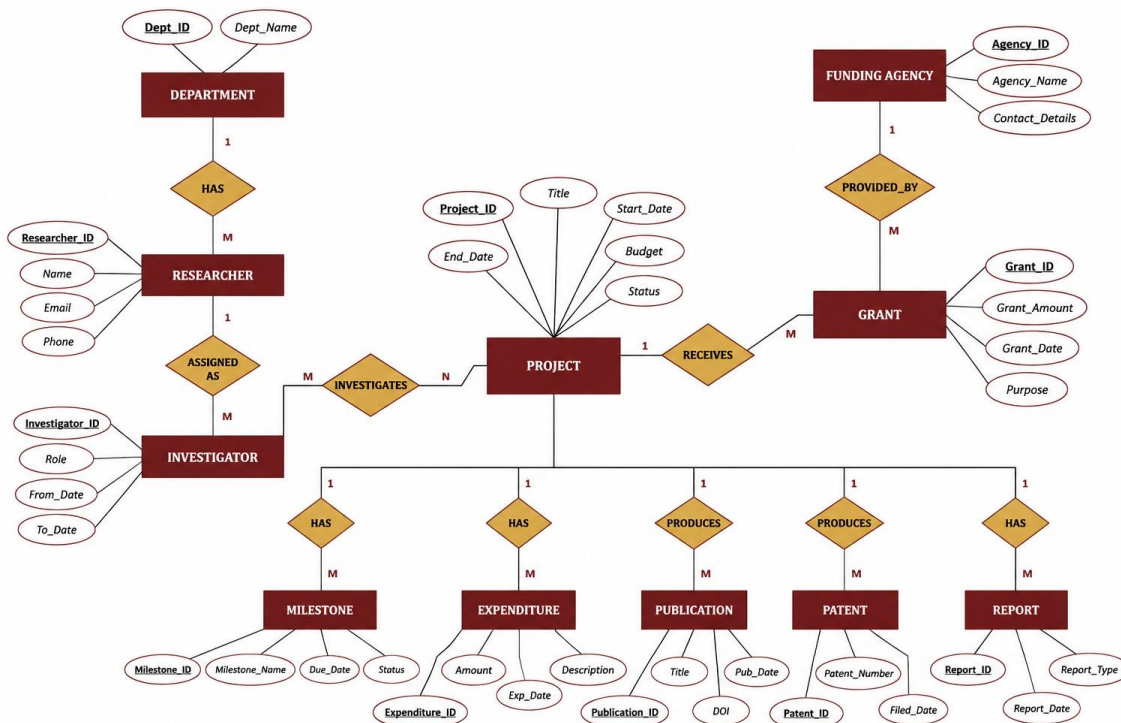


Figure 1. Conceptual ER diagram of the Research Project, Publication and Grant Management System.

5.1 Main Entities

Entity	Purpose
Department	Stores academic/research departments
Researcher	Stores researcher information
Project	Stores research project details
Investigator	Represents researcher involvement in projects
Funding Agency	Stores organizations providing funding
Grant	Stores project funding details
Milestone	Tracks project milestones
Expenditure	Records project expenses
Publication	Stores research publications
Patent	Stores project-related patents

Entity	Purpose
Report	Stores project/research reports

5.2 Relationships and Cardinality

Relationship	Entities	Cardinality
HAS	Department – Researcher	1 : M
ASSIGNED AS	Researcher – Investigator	1 : M
INVESTIGATES	Investigator – Project	M : N
RECEIVES	Project – Grant	1 : M
PROVIDED_BY	Funding Agency – Grant	1 : M
HAS	Project – Milestone	1 : M
HAS	Project – Expenditure	1 : M
PRODUCES	Project – Publication	1 : M
PRODUCES	Project – Patent	1 : M
HAS	Project – Report	1 : M

6. Relational Schema and Normalization

6.1 Relational Schema

Relation	Attributes
DEPARTMENT	Dept_ID (PK), Dept_Name
RESEARCHER	Researcher_ID (PK), Dept_ID (FK), Name, Email, Phone
PROJECT	Project_ID (PK), Title, Start_Date, End_Date, Budget, Status
INVESTIGATOR	Investigator_ID (PK), Researcher_ID (FK), Project_ID (FK), Role, From_Date, To_Date
FUNDING AGENCY	Agency_ID (PK), Agency_Name, Contact_Details

Relation	Attributes
RESEARCH GRANT	Grant_ID (PK), Project_ID (FK), Agency_ID (FK), Grant_Amount, Grant_Date, Purpose
MILESTONE	Milestone_ID (PK), Project_ID (FK), Milestone_Name, Due_Date, Status
EXPENDITURE	Expenditure_ID (PK), Project_ID (FK), Amount, Exp_Date, Description
PUBLICATION	Publication_ID (PK), Project_ID (FK), Title, Pub_Date, DOI
PATENT	Patent_ID (PK), Project_ID (FK), Patent_Number, Filed_Date
REPORT	Report_ID (PK), Project_ID (FK), Report_Type, Report_Date

The INVESTIGATOR relation acts as the link between RESEARCHER and PROJECT, resolving the many-to-many INVESTIGATES relationship and carrying the role and the period of involvement.

6.2 Foreign Key Summary

Relation	Foreign Key	References
RESEARCHER	Dept_ID	DEPARTMENT
INVESTIGATOR	Researcher_ID	RESEARCHER
INVESTIGATOR	Project_ID	PROJECT
RESEARCH GRANT	Project_ID	PROJECT
RESEARCH GRANT	Agency_ID	FUNDING AGENCY
MILESTONE	Project_ID	PROJECT
EXPENDITURE	Project_ID	PROJECT
PUBLICATION	Project_ID	PROJECT
PATENT	Project_ID	PROJECT
REPORT	Project_ID	PROJECT

6.3 Normalization

The relations are structured to avoid repeating groups and unnecessary duplication. Every table has a single-column primary key, so no partial dependencies exist. Non-key attributes describe only the entity

identified by the primary key, which supports Third Normal Form (3NF). Department names are stored only in DEPARTMENT, funding agency details only in FUNDING AGENCY, and project outputs (publications, patents, reports) and financial details (grants, expenditure) are separated into their own relations.

Normal Form	Applied Principle
1NF	Atomic attributes and no repeating groups.
2NF	Single-column primary keys remove partial dependency concerns.
3NF	Non-key attributes depend on the key and are separated by entity.

7. Data Dictionary

The data types below are the proposed types for implementation.

7.1 DEPARTMENT

Column	Type	Constraint / Meaning
Dept_ID	INT	PK; department identifier
Dept_Name	VARCHAR(100)	Department name

7.2 RESEARCHER

Column	Type	Constraint / Meaning
Researcher_ID	INT	PK; researcher identifier
Dept_ID	INT	FK → DEPARTMENT
Name	VARCHAR(100)	Researcher name
Email	VARCHAR(100)	Email address
Phone	VARCHAR(15)	Contact number

7.3 PROJECT

Column	Type	Constraint / Meaning
Project_ID	INT	PK; project identifier

Column	Type	Constraint / Meaning
Title	VARCHAR(200)	Project title
Start_Date	DATE	Project start date
End_Date	DATE	Project end date
Budget	DECIMAL(12,2)	Approved project budget
Status	VARCHAR(30)	Project status (e.g., Ongoing / Completed)

7.4 INVESTIGATOR

Column	Type	Constraint / Meaning
Investigator_ID	INT	PK
Researcher_ID	INT	FK → RESEARCHER
Project_ID	INT	FK → PROJECT
Role	VARCHAR(50)	Role in the project
From_Date	DATE	Start of involvement
To_Date	DATE	End of involvement

7.5 FUNDING AGENCY

Column	Type	Constraint / Meaning
Agency_ID	INT	PK
Agency_Name	VARCHAR(150)	Funding organization name
Contact_Details	VARCHAR(200)	Contact information

7.6 RESEARCH GRANT

Column	Type	Constraint / Meaning
Grant_ID	INT	PK
Project_ID	INT	FK → PROJECT
Agency_ID	INT	FK → FUNDING AGENCY
Grant_Amount	DECIMAL(12,2)	Amount sanctioned

Column	Type	Constraint / Meaning
Grant_Date	DATE	Date of grant
Purpose	VARCHAR(200)	Purpose of the grant

7.7 MILESTONE

Column	Type	Constraint / Meaning
Milestone_ID	INT	PK
Project_ID	INT	FK → PROJECT
Milestone_Name	VARCHAR(150)	Milestone description
Due_Date	DATE	Target completion date
Status	VARCHAR(30)	Pending / In Progress / Completed

7.8 EXPENDITURE

Column	Type	Constraint / Meaning
Expenditure_ID	INT	PK
Project_ID	INT	FK → PROJECT
Amount	DECIMAL(12,2)	Amount spent
Exp_Date	DATE	Date of expense
Description	VARCHAR(200)	Nature of expense

7.9 PUBLICATION

Column	Type	Constraint / Meaning
Publication_ID	INT	PK
Project_ID	INT	FK → PROJECT
Title	VARCHAR(250)	Publication title
Pub_Date	DATE	Publication date
DOI	VARCHAR(100)	Digital Object Identifier

7.10 PATENT

Column	Type	Constraint / Meaning
Patent_ID	INT	PK
Project_ID	INT	FK → PROJECT
Patent_Number	VARCHAR(50)	Patent number
Filed_Date	DATE	Date of filing

7.11 REPORT

Column	Type	Constraint / Meaning
Report_ID	INT	PK
Project_ID	INT	FK → PROJECT
Report_Type	VARCHAR(50)	Type of report
Report_Date	DATE	Date of report

8. SQL Commands (DDL and DML)

8.1 DDL

Representative DDL used to define the relational structure:

```
CREATE DATABASE research_management_db;
USE research_management_db;

CREATE TABLE department (
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_name VARCHAR(100) NOT NULL
);

CREATE TABLE researcher (
    researcher_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_id INT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    phone VARCHAR(15),
    FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);

CREATE TABLE project (
    project_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
title VARCHAR(200) NOT NULL,  
start_date DATE,  
end_date DATE,  
budget DECIMAL(12,2),  
status VARCHAR(30)  
);
```

8.2 DML

```
INSERT INTO department (dept_name)  
VALUES ('Computer Science and Engineering');  
  
INSERT INTO researcher (dept_id, name, email, phone)  
VALUES (1, 'Sample Researcher', 'researcher@example.com', '9876543210');  
  
UPDATE milestone  
SET status = 'Completed'  
WHERE milestone_id = 3;  
  
DELETE FROM expenditure  
WHERE expenditure_id = 10;
```

9. Sample Queries

9.1 Projects with their investigators

```
SELECT p.title, r.name, i.role  
FROM project p  
JOIN investigator i ON i.project_id = p.project_id  
JOIN researcher r ON r.researcher_id = i.researcher_id  
ORDER BY p.title;  
Purpose: lists each project with the researchers assigned to it and their roles.
```

9.2 Total grant funding per project

```
SELECT p.title, SUM(g.grant_amount) AS total_grant  
FROM project p  
JOIN research_grant g ON g.project_id = p.project_id  
GROUP BY p.project_id, p.title;  
Purpose: shows how much funding each project has received.
```

9.3 Presentation-II Query — Who and How many Researchers are working on the healthcare project?

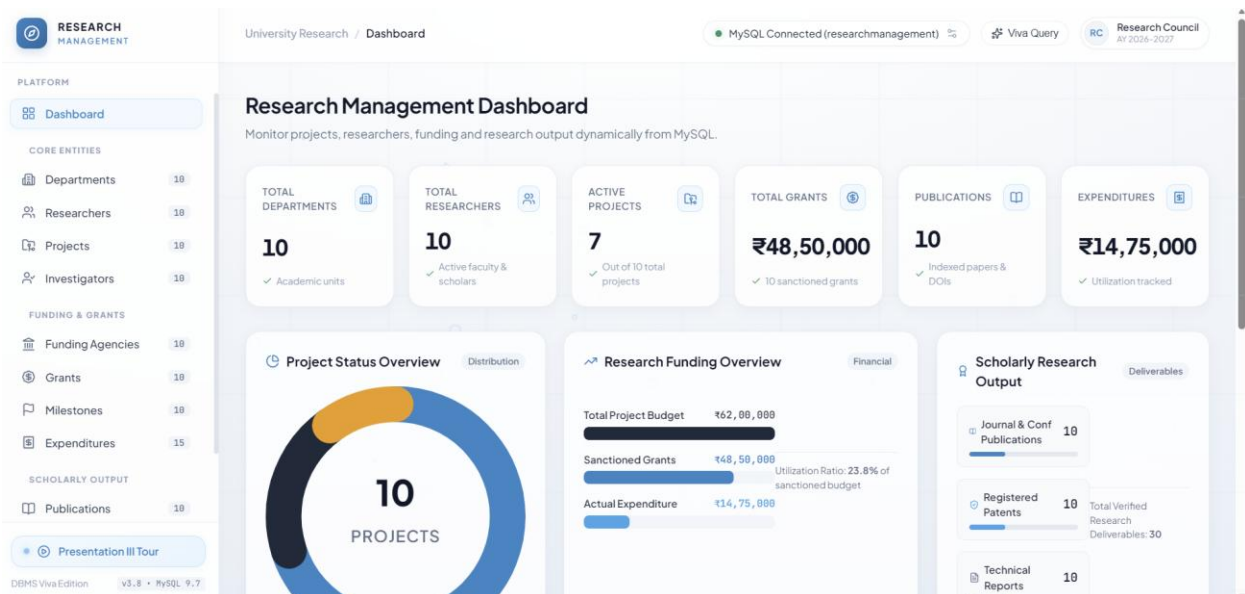
```
SELECT R.Name, COUNT(*) OVER () AS Number_of_Researchers  
FROM investigator I  
JOIN researcher R ON I.Researcher_ID = R.Researcher_ID
```

JOIN project P ON I.Project_ID = P.Project_ID
WHERE P.Title = 'AI Based Healthcare Analysis';

Name	Number_of_Researchers
Aarav Sharma	2
Priya Rao	2

10. UI Design & Screenshots

Dashboard Screenshot



Department Screenshot

RESEARCH MANAGEMENT

University Research / Departments

MySQL Connected (researchmanagement) Viva Query Research Council AR 2026-2027

PLATFORM

DASHBOARD

CORE ENTITIES

Departments 10

Researchers 10

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Investigators 10

FUNDING & GRANTS

Funding Agencies 10

Grants 10

Milestones 10

Expenditures 15

SCHOLARLY OUTPUT

Publications 10

Patents 10

Reports 10

Presentation III Tour

DBMS Viva Edition v3.8 - MySQL 9.7

Departments

Live MySQL records in researchmanagement.department

Search department records...

Total Records: 10 rows

DEPT ID PK	DEPT NAME	ACTIONS
#6	Artificial Intelligence	 
#7	Biomedical Engineering	 
#2	Biotechnology	 
#1	Computer Science	 
#5	Data Science	 
#4	Electrical Engineering	 
#8	Environmental Science	 
#9	Information Technology	 
#3	Mechanical Engineering	 
#10	Robotics Engineering	 

Researcher Screenshot

RESEARCH MANAGEMENT

University Research / Researchers

MySQL Connected (researchmanagement) Viva Query Research Council AR 2026-2027

PLATFORM

DASHBOARD

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Researchers 10

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Grants 10

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Expenditures 15

SCHOLARLY OUTPUT

Publications 10

Patents 10

Reports 10

Presentation III Tour

DBMS Viva Edition v3.8 - MySQL 9.7

Researchers

Live MySQL records in researchmanagement.researcher

Search researcher records...

Total Records: 10 rows

RESEARCHER ID PK	DEPT ID FK	NAME	EMAIL	PHONE	ACTIONS
#101	#1	Aarav Sharma	aarav@university.edu	9876543210	 
#102	#1	Priya Rao	priya@university.edu	9876543211	 
#103	#2	Rahul Mehta	rahul@university.edu	9876543212	 
#104	#3	Ananya Iyer	ananya@university.edu	9876543213	 
#105	#5	Vikram Singh	vikram@university.edu	9876543214	 
#106	#1	Madhu	madhu@university.edu	9876543217	 
#107	#7	Arijun Reddy	arjun@university.edu	9876543216	 
#108	#8	Sneha Patel	sneha@university.edu	9876543215	 
#109	#9	Karan Malhotra	karan@university.edu	9876543218	 
#110	#10	Meera Nair	meera@university.edu	9876543219	 

Projects Screenshot

RESEARCH MANAGEMENT

University Research / Projects

MySQL Connected (researchmanagement) Viva Query Research Council AY 2026-2027

PLATFORM

Dashboard

CORE ENTITIES

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SCHOLARLY OUTPUT

Publications 10

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Reports 10

Presentation III Tour

DBMS Viva Edition v3.8 MySQL 9.7

Projects

Live MySQL records in researchmanagement.project

Export CSV Add Record

Search project records...

Total Records: 10 rows

PROJECT ID PK	TITLE	START DATE	END DATE	BUDGET	STATUS	ACTIONS
#201	AI Based Healthcare Analysis	10 Jan 2026	20 Dec 2026	₹5,00,000	Active	
#202	Smart Agriculture System	15 Feb 2026	30 Oct 2026	₹3,00,000	Active	
#203	Sustainable Energy Optimization	1 Mar 2026	30 Nov 2026	₹7,50,000	Completed	
#204	Autonomous Vehicle Monitoring	20 Jan 2026	30 Sept 2026	₹4,50,000	Completed	
#205	Natural Language Processing for Education	1 Apr 2026	31 Mar 2027	₹9,00,000	Active	
#206	AI Powered Disease Detection	1 May 2026	30 Apr 2027	₹6,50,000	Active	
#207	Biomedical Signal Processing	1 Jun 2026	31 May 2027	₹5,50,000	Active	
#208	Climate Change Prediction Model	1 Jul 2026	30 Jun 2027	₹7,00,000	Active	
#209	Cybersecurity Threat Detection	15 May 2026	31 Mar 2027	₹6,00,000	Active	
#210	Robotic Assistance System	1 Aug 2026	31 Jul 2027	₹8,00,000	Pending	

Investigators Screenshot

RESEARCH MANAGEMENT

University Research / Investigators

MySQL Connected (researchmanagement) Viva Query Research Council AY 2026-2027

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Presentation III Tour

DBMS Viva Edition v3.8 MySQL 9.7

Investigators

Live MySQL records in researchmanagement.investigator

Export CSV Add Record

Search investigator records...

Total Records: 10 rows

INVESTIGATOR ID PK	RESEARCHER ID FK	PROJECT ID FK	ROLE	FROM DATE	TO DATE	ACTIONS
#401	#101	#201	Lead Investigator	10 Jan 2026	—	
#402	#102	#201	Co-Investigator	15 Jan 2026	—	
#403	#103	#202	Lead Investigator	15 Feb 2026	—	
#404	#104	#203	Lead Investigator	1 Mar 2026	30 Nov 2026	
#405	#105	#205	Lead Investigator	1 Apr 2026	—	
#406	#106	#206	Lead Investigator	1 May 2026	—	
#407	#107	#207	Lead Investigator	1 Jun 2026	—	
#408	#108	#208	Co-Investigator	1 Jul 2026	—	
#409	#109	#209	Lead Investigator	15 May 2026	—	
#410	#110	#210	Lead Investigator	1 Aug 2026	—	

Funding Agencies Screenshot

RESEARCH MANAGEMENT

University Research / Funding Agencies

MySQL Connected (researchmanagement) Viva Query Research Council AY 2026-2027

Funding Agencies

Live MySQL records in researchmanagement.funding_agency

Search funding_agency records...

Total Records: 10 rows

AGENCY ID PK	AGENCY NAME	CONTACT DETAILS	ACTIONS
#301	Department of Science and Technology	dst@gov.in	
#302	Indian Council of Medical Research	icmr@gov.in	
#303	University Research Fund	research@university.edu	
#304	Ministry of Education	education@gov.in	
#305	National Research Foundation	nrf@gov.in	
#306	Indian Institute of Science Research Fund	research@isc.ac.in	
#307	Department of Biotechnology	dbt@gov.in	
#308	Indian Space Research Organisation	research@isro.gov.in	
#309	National Informatics Centre	research@nic.in	
#310	Council of Scientific and Industrial Research	csir@gov.in	

DBMS Viva Edition v3.8 • MySQL 9.7

Grants Screenshot

RESEARCH MANAGEMENT

University Research / Research Grants

MySQL Connected (researchmanagement) Viva Query Research Council AY 2026-2027

Research Grants

Live MySQL records in researchmanagement.research_grant

Search research_grant records...

Total Records: 10 rows

GRANT ID PK	PROJECT ID FK	AGENCY ID FK	GRANT AMOUNT	GRANT DATE	PURPOSE	ACTIONS
#501	#201	#302	₹4,00,000	15 Jan 2026	Healthcare AI Research	
#502	#202	#301	₹2,50,000	20 Feb 2026	Smart Agriculture Research	
#503	#203	#303	₹6,00,000	5 Mar 2026	Sustainable Energy Research	
#504	#204	#304	₹3,50,000	1 Feb 2026	Autonomous Vehicle Research	
#505	#205	#305	₹7,00,000	10 Apr 2026	Educational AI Research	
#506	#206	#306	₹5,00,000	10 May 2026	AI Disease Detection Research	
#507	#207	#307	₹4,00,000	15 Jun 2026	Biomedical Signal Research	
#508	#208	#308	₹5,50,000	10 Jul 2026	Climate Prediction Research	
#509	#209	#309	₹4,50,000	25 May 2026	Cybersecurity Research	
#510	#210	#310	₹6,50,000	10 Aug 2026	Robotics Research	

DBMS Viva Edition v3.8 • MySQL 9.7

Milestone Screenshot

RESEARCH MANAGEMENT

University Research / Milestones

MySQL Connected (researchmanagement) Viva Query Research Council Apr 2026 - 2027

Export CSV + Add Record

Milestones

Live MySQL records in researchmanagement.milestone

Search milestone records...

Total Records: 10 rows

milestone id	Project ID	Milestone Name	Due Date	Status	Actions
#601	#201	Dataset Collection	15 Apr 2026	Completed	
#602	#202	Prototype Development	15 Jul 2026	In Progress	
#603	#203	System Testing	15 Oct 2026	Pending	
#604	#204	Vehicle Model Testing	20 Aug 2026	Completed	
#605	#205	NLP Model Training	15 Dec 2026	In Progress	
#606	#206	Medical Dataset Preparation	15 Sept 2026	Completed	
#607	#207	Signal Data Collection	15 Oct 2026	In Progress	
#608	#208	Climate Dataset Analysis	20 Nov 2026	Pending	
#609	#209	Threat Model Development	30 Oct 2026	In Progress	
#610	#210	Robot Prototype Design	15 Dec 2026	Pending	

DBMS Viva Edition v3.8 - MySQL 9.7

Expenditure Screenshot

RESEARCH MANAGEMENT

University Research / Expenditures

MySQL Connected (researchmanagement) Viva Query Research Council Apr 2026 - 2027

Export CSV + Add Record

Expenditures

Live MySQL records in researchmanagement.expenditure

Search expenditure records...

Total Records: 15 rows

Expenditure ID	Project ID	Amount	Exp Date	Description	Actions
#701	#201	₹1,20,000	20 Apr 2026	Research Equipment	
#702	#201	₹50,000	10 May 2026	Software License	
#703	#202	₹80,000	10 Jun 2026	Agricultural Sensors	
#704	#202	₹40,000	5 Jul 2026	Software and Materials	
#705	#203	₹1,50,000	5 Aug 2026	Laboratory Equipment	
#706	#203	₹75,000	25 Aug 2026	Energy Testing Materials	
#707	#204	₹1,00,000	20 May 2026	Vehicle Sensors	
#708	#204	₹60,000	15 Jun 2026	Testing Equipment	
#709	#205	₹2,00,000	10 Aug 2026	Computing Resources	
#710	#205	₹1,00,000	1 Sept 2026	Dataset and Software	

DBMS Viva Edition v3.8 - MySQL 9.7

Publications Screenshot

RESEARCH MANAGEMENT

University Research / Publications

MySQL Connected (researchmanagement) Viva Query Research Council AY 2026-2027

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Presentation III Tour

DBMS Viva Edition v3.8 • MySQL 9.7

Publications

Live MySQL records in researchmanagement.publication

Search publication records...

Total Records: 18 rows

PUBLICATION ID FK	PROJECT ID FK	TITLE	PUB DATE	DOI	ACTIONS
#801	#201	AI-Based Analysis of Medical Images	10 May 2026	10.1000/aihealth001	 
#802	#202	Machine Learning for Smart Agriculture	20 Jul 2026	10.1000/smartagri002	 
#803	#203	AI Optimization for Sustainable Energy	1 Sept 2026	10.1000/energy003	 
#804	#204	Autonomous Vehicle Monitoring Using AI	15 Aug 2026	10.1000/autoveh004	 
#805	#205	Natural Language Processing in Education	10 Sept 2026	10.1000/nlpedu005	 
#806	#206	AI Based Disease Detection Using Deep Learning	20 Sept 2026	10.1000/disease006	 
#807	#207	Biomedical Signal Processing Using Machine Learning	5 Oct 2026	10.1000/biomed007	 
#808	#208	Machine Learning for Climate Prediction	15 Oct 2026	10.1000/climate008	 
#809	#209	AI Based Cybersecurity Threat Detection	30 Sept 2026	10.1000/cyber009	 
#810	#210	Intelligent Robotic Assistance Systems	10 Nov 2026	10.1000/robot010	 

Patents Screenshot

RESEARCH MANAGEMENT

University Research / Patents

MySQL Connected (researchmanagement) Viva Query Research Council AY 2026-2027

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Patents

Live MySQL records in researchmanagement.patent

Search patent records...

Total Records: 18 rows

PATENT ID FK	PROJECT ID FK	PATENT NUMBER	FILED DATE	ACTIONS
#901	#201	PAT-AI-001	1 Jun 2026	 
#902	#202	PAT-AGR-002	10 Aug 2026	 
#903	#203	PAT-ENE-003	5 Sept 2026	 
#904	#204	PAT-AUTO-004	12 Jul 2026	 
#905	#205	PAT-NLP-005	12 Sept 2026	 
#906	#206	PAT-DIS-006	1 Oct 2026	 
#907	#207	PAT-BIO-007	10 Oct 2026	 
#908	#208	PAT-CLI-008	1 Nov 2026	 
#909	#209	PAT-CYB-009	5 Oct 2026	 
#910	#210	PAT-ROB-010	15 Nov 2026	 

Reports Screenshot

The screenshot displays the 'Reports' section of a 'RESEARCH MANAGEMENT' system. The interface includes a sidebar with navigation links for Dashboard, Core Entities (Departments, Researchers, Projects, Investigators), Funding & Grants (Funding Agencies, Grants, Milestones, Expenditures), and Scholarly Output (Publications, Patents, Reports). The main content area shows a table of reports with columns for Report ID, Project ID, Report Type, Report Date, and Actions. A search bar and 'Total Records: 10 rows' indicator are also present.

REPORT ID	PROJECT ID	REPORT TYPE	REPORT DATE	ACTIONS
#1001	#201	Project Progress Report	30 Jun 2026	
#1002	#202	Funding Utilization Report	31 Jul 2026	
#1003	#203	Final Research Report	10 Sept 2026	
#1004	#204	Project Completion Report	15 Sept 2026	
#1005	#205	Research Progress Report	20 Sept 2026	
#1006	#206	Medical AI Progress Report	1 Oct 2026	
#1007	#207	Biomedical Research Report	15 Oct 2026	
#1008	#208	Climate Research Progress Report	1 Nov 2026	
#1009	#209	Cybersecurity Project Report	20 Oct 2026	
#1010	#210	Robotics Development Report	20 Nov 2026	

10. Conclusion

The Research Project, Publication and Grant Management System provides a complete relational design for managing research activity. The eleven-entity model keeps people, projects, funding, progress and research outputs in separate but linked tables, and foreign keys preserve the relationships between them. The design supports project tracking, grant and budget monitoring, milestone follow-up and research-output reporting from one consistent database.

11. Appendix: GitHub Repository

The complete project source code, frontend, backend structure, startup scripts, and project documentation are maintained in the following repository:

<https://github.com/cc-91/DBMS-Course-Project>

Repository contents:

- Project Presentations (PPTs)
- Project Report